Akoizo Original Practice - Form 1 - Answer key

Division B | Solar System | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) Which three broad needs define habitability for life as we know it?

A. Liquid water, energy, and key chemical elements

B. Oxygen, sunlight, and soil

C. A magnetic field, rings, and moons

D. Ice, methane, and nitrogen only

Correct: A

2. (2 points) Why is Venus's surface hotter than Mercury's?

Unit: Venus and Mars: rocky neighbors.
Worked-case data: A Mars orbiter detects clay minerals in layered rocks at the end of a channel that opens into a crater, forming a fan shape. What can you conclude?
Separate investigation observations:
Orbital image: A fan of layered sediment where a valley enters a crater.
Spectra: Clay minerals and carbonates detected in the fan.
Rover data: Fine-grained mudstone layers on the crater floor.

A. A thick CO_2 atmosphere creates a runaway greenhouse

B. Venus is closer to the Sun

C. Venus has a strong magnetic field

D. Venus has active tidal heating

Correct: A

3. (2 points) Which method directly gives a planet's radius?

Unit: Exoplanets: detection and the named systems.
Worked-case data: An Earth-sized planet transits two stars: a Sun-like star and a red dwarf with one-tenth the Sun's radius. Compare the transit depths.
Separate investigation observations:
Light curve: Seven different periodic dips in brightness.
Star: An ultracool red dwarf, about 40 light-years away.
Sizes: All planets are roughly Earth-sized.
JWST: The two innermost planets show no thick atmosphere.

A. Transit

B. Radial velocity

C. Microlensing alone

D. Astrometry

Correct: A

4. (2 points) If the distance between two masses triples, the gravitational force becomes:

A. 3 times as large

B. 1/27 as large

C. 1/9 as large

D. 1/3 as large

Correct: C

5. (2 points) Which class of molecules forms cell membranes?

Unit: Small bodies and the building blocks of life.
Worked-case data: Earth's oceans have a D/H ratio of about 1.6 x 10□^4. Comet 67P's water has about 5.3 x 10□^4, and carbon-rich asteroids have values close to Earth's. Which source better matches Earth's water?
Separate investigation observations:
Sample minerals: Clay minerals and sodium carbonate salts.
Organics: Amino acids and nucleobases detected in the lab.
Isotopes: D/H ratio close to Earth's oceans.

A. Nucleic acids

B. Carbohydrates

C. Lipids

D. Proteins

Correct: C

6. (2 points) A cool object near 300 K emits most strongly in:

A. Infrared

B. Visible

C. Ultraviolet

D. X-rays

Correct: A

7. (2 points) What liquids fill Titan’s lakes and seas?

Unit: Ocean worlds: Europa, Ganymede, Enceladus, and Titan.
Worked-case data: A spacecraft flying by a moon detects a magnetic field that changes direction as the moon moves through its planet’s field. What does this suggest about the moon’s interior?
Separate investigation observations:
Atmosphere: Thick nitrogen, 1.5 times Earth’s surface pressure.
Surface temperature: About -180 °C.
Radar: Smooth, dark lakes near the poles.
Probe images: Rounded pebbles on the landing site.

A. Methane and ethane

B. Liquid water

C. Sulfuric acid

D. Liquid nitrogen only

Correct: A

8. (4 points) Explain why being in the habitable zone does not guarantee liquid water.

answer: Surface temperature also depends on albedo and atmosphere. A planet with no atmosphere may freeze, and one with a runaway greenhouse, like Venus, may be too hot. (4 points).

9. (4 points) Explain why liquid water is not stable on Mars’s surface today.

Unit: Venus and Mars: rocky neighbors.
Worked-case data: A Mars orbiter detects clay minerals in layered rocks at the end of a channel that opens into a crater, forming a fan shape. What can you conclude?
Separate investigation observations:
Orbital image: A fan of layered sediment where a valley enters a crater.
Spectra: Clay minerals and carbonates detected in the fan.
Rover data: Fine-grained mudstone layers on the crater floor.

answer: The atmospheric pressure is so low that water quickly boils or sublimes, and temperatures are usually below freezing. (4 points).

10. (4 points) Explain how combining transit and radial velocity data reveals a planet's composition.

Unit: Exoplanets: detection and the named systems.
Worked-case data: An Earth-sized planet transits two stars: a Sun-like star and a red dwarf with one-tenth the Sun's radius. Compare the transit depths.
Separate investigation observations:
Light curve: Seven different periodic dips in brightness.
Star: An ultracool red dwarf, about 40 light-years away.
Sizes: All planets are roughly Earth-sized.
JWST: The two innermost planets show no thick atmosphere.

answer: Transits give the radius and radial velocity gives the mass. Together they give density, which shows whether the planet is rocky, water-rich, or gaseous. (4 points).

11. (4 points) Why does the Moon cause larger tides on Earth than the Sun?

Unit: Gravity, orbits, and tides.
Worked-case data: Planet A has 4 times Earth's mass and 2 times Earth's radius. Compare its surface gravity and escape velocity with Earth's.
Separate investigation observations:
Orbit: Moon B orbits its planet in 3.6 days on a slightly elliptical path.
Rotation: It always shows the same face to the planet.
Neighbors: Its period is exactly twice that of an inner moon.
Surface: Young, crack-covered ice with few craters.

answer: Tidal force scales with M/d^3. The Moon is much closer, so the distance factor outweighs the Sun's much larger mass. (4 points).

12. (4 points) What is a clathrate, and where might it matter?

Unit: Small bodies and the building blocks of life.
Worked-case data: Earth's oceans have a D/H ratio of about 1.6 x 10□^4. Comet 67P's water has about 5.3 x 10□^4, and carbon-rich asteroids have values close to Earth's. Which source better matches Earth's water?
Separate investigation observations:
Sample minerals: Clay minerals and sodium carbonate salts.
Organics: Amino acids and nucleobases detected in the lab.
Isotopes: D/H ratio close to Earth's oceans.

answer: An ice structure that traps gas molecules like methane inside cages; it may store methane on Titan, Mars, or icy moons. (4 points).

13. (4 points) Star A is 6,000 K and star B is 3,000 K. Compare their peak wavelengths.

answer: λmax □ 1/T, so star B peaks at twice the wavelength of star A, farther toward the red and infrared. (4 points).

14. (4 points) Compare Titan’s methane cycle with Earth’s water cycle.

Unit: Ocean worlds: Europa, Ganymede, Enceladus, and Titan.
Worked-case data: A spacecraft flying by a moon detects a magnetic field that changes direction as the moon moves through its planet’s field. What does this suggest about the moon’s interior?
Separate investigation observations:
Atmosphere: Thick nitrogen, 1.5 times Earth’s surface pressure.
Surface temperature: About -180 °C.
Radar: Smooth, dark lakes near the poles.
Probe images: Rounded pebbles on the landing site.

answer: Methane evaporates, forms clouds, rains, flows in rivers, and collects in lakes, just as water does on Earth, but at much lower temperatures. (4 points).

15. (4 points) Why would oxygen and methane together in an exoplanet atmosphere be interesting?

answer: They react with each other, so both being present means something keeps resupplying them, which on Earth is life. (4 points).

16. (4 points) Why did Perseverance land in Jezero crater?

Unit: Venus and Mars: rocky neighbors.
Worked-case data: A Mars orbiter detects clay minerals in layered rocks at the end of a channel that opens into a crater, forming a fan shape. What can you conclude?
Separate investigation observations:
Orbital image: A fan of layered sediment where a valley enters a crater.
Spectra: Clay minerals and carbonates detected in the fan.
Rover data: Fine-grained mudstone layers on the crater floor.

answer: Jezero contains an ancient river delta and lake deposits, an environment that may have preserved signs of ancient microbial life. (4 points).

17. (4 points) Why is direct imaging easiest for young, large planets far from their stars?

answer: They are bright in infrared because they are still hot, and wide separation makes them easier to resolve next to the star’s glare. (4 points).

18. (6 points) Solve this case and explain your reasoning: Planet A has 4 times Earth’s mass and 2 times Earth’s radius. Compare its surface gravity and escape velocity with Earth’s.

reasoning: g □ M/R^2: 4 ÷ 2^2 = 1, so surface gravity equals Earth’s.
v_esc □ sqrt (M/R): sqrt (4 ÷ 2) = sqrt 2 about 1.4.
Earth’s v_esc is 11.2 km/s, so Planet A’s is about 15.8 km/s. (3 points).

conclusion: Planet A has the same surface gravity but about 1.4 times Earth’s escape velocity, helping it retain an atmosphere. (3 points).